Salahaddin University-Erbil
College of Computer Science and IT
Stage 2 Cs & IT

Web Technology Basics - Lec. 1 – CSS

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Tuesday: IT- 8:30-10:30/ Hall 3
CS- 10:30 -12:30/ Hall 16

What you will gain from this course:

- Style and layout pages with CSS.
- Make pages responsive, accessible, and performant.

Outline

- Cascading Style Sheet (CSS)
- CSS Advantages
- CSS Syntax
- Styling Methods
- Inheritance
- CSS Selectors
- CSS Text
- Operators
- Pseudoclasses
- Pseudoelements
- CSS Comments
- CSS Styles
  - ✓ Floating & Fixed Position
  - ✓ Borders
  - ✓ Boxes
  - ✓ Tables
  - ✓ Lists

Cascading Style Sheet

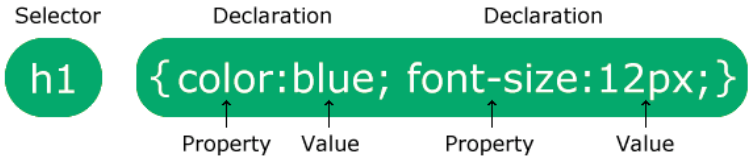
- Cascading Style Sheets, referred to as CSS, is a simple design language intended to simplify the process of making web pages presentable.
- Using CSS, you can control the color of the text, the style of fonts, the spacing between paragraphs, how columns are sized and laid out, what background images or colors are used, as well as a variety of other effects.
- CSS is easy to learn and understand but it provides powerful control over the presentation of an HTML document. Most commonly, CSS is combined with the markup languages HTML or XHTML.

CSS Advantages

- **CSS saves time -** You can write CSS once and then reuse same sheet in multiple HTML pages. You can define a style for each HTML element and apply it to as many Web pages as you want.
- **Pages load faster -** If you are using CSS, you do not need to write HTML tag attributes every time. Just write one CSS rule of a tag and apply to all the occurrences of that tag. So less code means faster download times.
- **Easy maintenance -** To make a global change, simply change the style, and all elements in all the web pages will be updated automatically.
- **Superior styles to HTML -** CSS has a much wider array of attributes than HTML so you can give far better look to your HTML page in comparison of HTML attributes.
- **Global web standards -** Now HTML attributes are being deprecated and it is being recommended to use CSS. So it’s a good idea to start using CSS in all the HTML pages to make them compatible to future browsers.

CSS Syntax

- A CSS rule consists of a **selector** and a **declaration** block.
- The **selector** points to the HTML element you want to style. This could be any tag like <h1> or <table> etc.
- The **declaration** block contains one or more declarations separated by *semicolons* and declaration blocks are surrounded by *curly braces*.
- Each declaration includes a CSS **property** name and a **value**, separated by a *colon*.
- A **property** is a type of attribute of HTML tag. The HTML attributes are converted into CSS properties. They could be color, border, …etc.
- **Values** are assigned to properties. For example color property can have value either red or #F1F1F1 etc.



Styling Methods

- There are three ways of inserting a style sheet:

1. Inline style

An inline style loses many of the advantages of style sheets by mixing content with presentation. To use inline styles you use the style attribute in the relevant tag. The style attribute can contain any CSS property. The example shows how to change the color and the left margin of a paragraph:

**e.g: This is a paragraph. **

Styling Methods

2. Internal Style Sheet

An internal style sheet should be used when a single document has a unique style. You define internal styles in the head section of an HTML page, by using the <style> tag, like this:

```
<head>
<style>
hr {color:sienna;}
p {margin-left:20px;}
body {background-image:url("images/back40.gif");}
</style>
</head>
```


Styling Methods

3. External Style Sheet:

An external style sheet is ideal when the style is applied to many pages. With an external style sheet, you can change the look of an entire website by changing one file. Each page must link to the style sheet using the <link> tag. The <link> tag goes inside the head section:

e.g:

```
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
</head>
```

Styling Methods

3. External Style Sheet:

- The rel attribute specifies the relationship between the current document and the linked document. Only used if the href attribute is present.
- An external style sheet can be written in any text editor. The file should not contain any html tags. Your style sheet should be saved with a .css extension. An example of a style sheet file is shown below:

e.g:

```
hr {color:sienna;}
p {margin-left:20px;}
body {background-image:url("images/back40.gif");}
```


Change heading & Paragraph Font Style

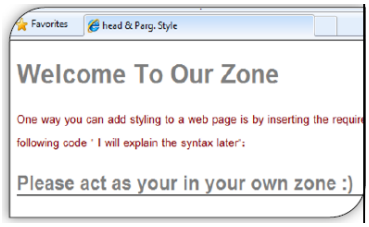
```
<!DOCTYPE html>
<html>
<head>
<title>HTML
Frames</title>
<style>
h1 {font-family: sans-serif;
color: gray;}
h2 {
font-family: sans-serif;
color: gray;
}
p {color: maroon; }
</style>
</head>
<body>
<h1> Welcome To Our Zone</h1>
<p>One way you can add styling to a web page
is by inserting the required statements into…
</p>
<h2> Please act as in your own zone :) </h2>
</body>
</html>
```

Adding a rule for more than one element

- To write a rule for more than one element, just put commas between the selectors, like “h1, h2”.

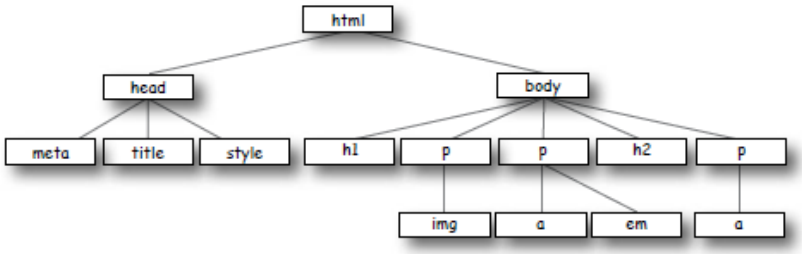
Example:

```
<!DOCTYPE html>
<html>
<head>
<title>head & Parg. Style</title>
<style>
h1,h2 {font-family: sans-serif;
color: gray; }
h2 {border-bottom:1px solid black; }
p {color: maroon; }
</style>
</head>
<body>
<h1> Welcome To Our Zone</h1>
<p>One way you can add styling to a web page is by inserting the required statements into… </p>
<h2> Please act as in your own zone :) </h2>
</body>
</html>
```



Inheritance

- Just like biological characteristics are inherited in human, elements can inherit styles from their parents. In this case, the <a> and elements inherited the font family style from the <p> element, which is their parent element.
- Note: Not every style is inherited. Just some are, like font-family.

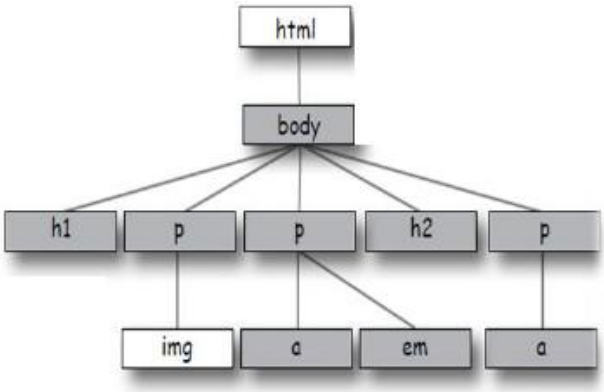


Inheritance Example

```
<html>
<head> <title>head & Parg. Style</title>
<style>
h1,h2 {font-family: sans-serif;color: gray; }
h2 {border-bottom:1px solid black; }
p {font-family: sans-serif; color: maroon; }
</style></head>
<body>
 <h1> Welcome to Hawler </h1>
 <p>  <img src="Penguins.jpg" alt="Drinks“ style="height:200px;width:200px" > </p>
 <p> Join us any evening for refreshing <a href="links.html">clear air</a>, conversation and maybe a game or two of
        <em> skiing on Korek Mountain</em>. Wireless access is always provided (Bring your own web server).
</p>
 <h2> Directions </h2>
 <p> You'll find us right in the center of downtown Webville. If you need help finding us, check out our
        <a   href="links.html"> detailed directions</a>. Come join us!  </p>
```

What if we move the font up the family tree?

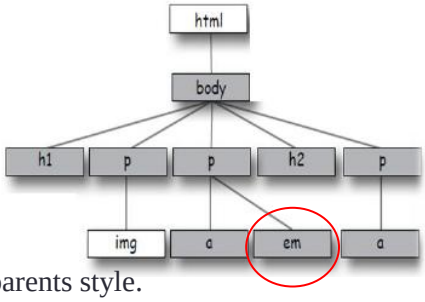
- That should have the effect of changing the font for all the <body> element's children, and children's children



Overriding Inheritance

- By moving the font-family property up into the body, you've set that font style for the entire page. But what if you don't want the sans-serif font on every element? For instance, you could decide that you want elements to use the serif font instead.

```
<style>
body {font-family: sans-serif;}
h1, h2 {color: gray;}
h1{border-bottom: 1px solid black;}
p{color: maroon;}
em{font-family: serif;}
</style>
```



- Assigning a style to specific element will override parents style.
- In this case assigning font family for em as serif will override the font family of the body, so the em element will have Serif font family.

CSS Selectors

- A CSS selector selects the HTML element(s) you want to style.

1. **Element Selector:** The element selector specifies the HTML element to style, such as or <i>.

For example, the following rule will ensure that all text within ... tags is fully justified

Example: p { text-align : justify; }

2. ID Selector: If you give an element an ID (like this:) , you can directly access it from CSS in the following way, which changes all the text in the named element to italic:

**** In css file or inside <style>****

```
#mydiv { font-style : italic; }
#myid { text-decoration : underline; }
```

**** In html file ****

```
<div id='myid'>Hello</div> <p id='myid'> welcome</p>
```

You can ensure that CSS applies the rule to both occurrences like this:

div#myid { text-decoration:underline; }

p#myid { text-decoration:underline; }

CSS Selectors

3. Class Selector: When there are several elements in a page that you want to share the same styling, you can assign them all the same class name, like this:

****In HTML file****

<h1 class='myclass'> <p class=‘myclass’>

and then create a single rule to modify all those elements at once, as in the following rule, which creates a 10-pixel left margin offset for all elements using the class:

**** In CSS file****

.myclass

{ margin-left : 10px; **}**

- You can narrow the scope of action of a class by specifying the types of elements to which it should apply. For example, the following rule applies the setting only to paragraphs that use the class main:

p.main

{ text-indent : 30px; }

CSS Selectors

4. Attribute Selector: Using this method you can directly reference attributes in the following manner, which sets all elements with the attribute type="submit" to a width of 100 pixels:

[type="submit"]

{

width : 100px;

}

- If you wish to narrow down the scope of the selector to, for example, only form input elements with that attribute type, you could use the following rule instead:

input[type="submit"]

{

width : 100px;

}

CSS Selectors

5. Universal Selector : The Universal Selector The " ***** " wildcard or universal selector matches any element, so the following rule will make a complete mess of a document by giving a green border to all of its elements:

```
*
{
border:1px solid green;
}
```

- It's unlikely that you will use the * on its own, but as part of a compound rule it can be very powerful. For example, the following rule will apply the same styling as above, but only to all paragraphs that are sub elements of the element with the ID aa, and only as long as they are not direct children:

```
#aa * p {border:1px solid green; }
```


CSS Selectors

6. Selecting by Group: With CSS it is possible to apply a rule to more than one element, class, or any other type of selector at the same time by separating the selectors with commas.

- For example, the following rule will place a dotted orange line underneath all paragraphs, the element with the ID of idname, and all elements using the class classname:

```
p, #idname, .classname
{
border-bottom:1px dotted orange;
}
```

CSS Text

1. **Text Color**: color property is used to set the color of the text. The color is specified by (color name, hex code or RGB value).

   Ex:h1{color:RGB(255,0,0);}

2. **Text Alignment**: text-align property is used to set the horizontal alignment of a text. A text can have *left, right, center or justified* alignment.

   Ex: body{ text-align : left; } p { text-align : justify; }

3. **Vertical align**: vertical-align property is used to set the vertical alignment of an element. The text can have *text-top, text-bottom, sub or super vertical* alignment.

   Ex:

   img { vertical-align : text-top; }

An image with a default alignment.

vertical-align: text-top:

An image with a text-top alignment.

vertical-align: text-bottom:

An image with a text-bottom alignment.

vertical-align: sub:

An image with a sub alignment.

vertical-align: sup:

An image with a super alignment.

CSS Text

4. **Text Align Last**: text-align-last property is used to set the horizontal alignment of the last line of a text. A text can have left, right, center or justified last line alignment.

   Ex: body { text-align-last : left; } p { text-align-last : justify; }

5. **Text Direction**: direction is used to change the text direction of an element.

   Ex: p { direction : rtl; }

6. **Text Decoration**: text-decoration property is used to specify the decoration added to text. It's a shorthand of text-decoration-line, text-decoration-color and text-decoration-style.

   text-decoration syntax:

   text-decoration: text-decoration-line text-decoration-color text-decoration-style |initial |inherit;

   text-decoration-line: underline|overline| line-through|inherit.

   text-decoration-color: color name, RGB, Hex code;

   text-decoration-style: solid, wavy, dotted, dashed or double;

   Ex: h1{text-decoration: underline overline red wavy;}

Text-decoration effect!

CSS Text

7. **Text Transform**: text-transform property is used to control the capitalize of text.

   text-transform:none|capitalize|uppercase|lowercase|initial|inherit;

   Ex: h1{ text-transform : capitalize; }

8. **Text Indentation**: text-indent property is used to specify the indentation of the first line of a text.

   text-indent :length|initial|inherit;

   Ex: p { text-indent : 30px; } or 10% or -2em

9. **Letter Spacing**: letter-spacing property increases or decreases the space between characters in a text.

   Letter-spacing: normal| length| initial |inherit;

   Ex: h3 { letter-spacing : 3px; } or -1px

CSS Text

10. **Line Height**: line-height specifies the height of a line.

    Syntax : Line-height:normal|number|length|initial|inherit;

    Ex: div { line-height : normal; } or 1.6 or 80%

11. **Word Spacing**: word-spacing property increases or decreases the white space between words.

    Syntax: word-spacing:normal|number|length|initial|inherit;

    Ex: p { word-spacing : 10%; } or 30px or normal

12. **White Space**: white-space property specifies how white-space inside an element is handled.

    Syntax: white-space:normal| nowrap| pre |initial |inherit;

    Ex: p { white-space : nowrap; }

CSS Text

13. Text Shadow: text-shadow property adds shadow to text. This property accepts a comma-separated list of shadows to be applied to the text.

Syntax: text-shadow: h-shadow v-shadow blur-radius color| none| initial| inherit;

Where: h-shadow is the position of the horizontal shadow and accepts negative values, while v-shadow is the position of vertical shadow and accepts negative values

Ex1: h2 { text-shadow : 2px 2px #ff0000; }

Ex2: h1 { color : white; text-shadow : 5px 5px 5px red, 10px 10px 5px blue,15px 15px 5px black; }

CSS Text

14. **Font type**: font-family property to specify the font of a text. The font-family property can hold several font names as a "fallback" system. If the browser does not support the first font, it tries the next font.

    Ex: .p1 { font-family : "Times New Roman", Times, serif; }

15. **Font Size**: The font-size property sets the size of a font.

    Syntax:font-size:medium|xx-small|x-small|small|large|x-large|xx-large|smaller|larger|length|initial|inherit;

    Ex: div.a { font-size : 15px; }

16. **Font Style**: The font-style property specifies the font style for a text.

    Syntax: font-style: normal|italic|oblique|initial|inherit;

    Ex: div.a { font-style : normal; }

17. Font weight: font-weight property sets how thick or thin characters in text should be displayed.

Syntax: font-weight: normal|bold|bolder|lighter|number|initial|inherit;

Note: Number Defines from thin to thick characters. 400 is the same as normal, and 700 is the same as bold

18. Font variant: The font-variant property specifies whether or not a text should be displayed in a small-caps font.

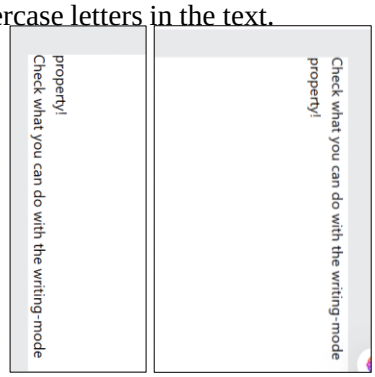
- In a small-caps font, all lowercase letters are converted to uppercase letters. However, the converted uppercase letters appears in a smaller font size than the original uppercase letters in the text.

Syntax: font-variant: normal|small-caps|initial|inherit;

19. font shorthand

font: italic small-caps bold 12px/30px Georgia, serif;

20. Writing-mode : Specify whether lines of text are laid out horizontally or vertically.

Syntax: writing-mode: horizontal-tb|vertical-rl|vertical-lr;

p { writing-mode : vertical-rl; }